Exploratory Data Analysis Report

Phishing URL Detection Dataset — Week 2 Assignment

INTERN DETAILS

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GitHub	Link

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1. Dataset Overview

Property	Value
Total Records	11,430
Total Features	89 (88 predictors + 1 target)
Target Column	`status` (legitimate / phishing)
Missing Values	**0** (no missing data)
Duplicate Rows	Not detected
Feature Types	2 string columns (`url`, `status`), 87 integer/float columns

The dataset is a rich collection of URL-based, HTML content-based, and external service-based features extracted from web pages, intended for phishing detection.

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2. Target Variable — Class Balance

Class	Count	Proportion
`legitimate`	5,715	50.0%
`phishing`	5,715	50.0%

Insight: The dataset is **perfectly balanced** (50/50 split). This is ideal for classification — no resampling techniques (SMOTE, oversampling) are needed. Model performance can be evaluated directly with accuracy, F1-score, and AUC-ROC without class-weight adjustments.

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3. Feature Categories

Features were grouped into four logical categories:

3.1 URL-Structure Features (lexical)

These capture statistical properties of the raw URL string: `length_url`, `length_hostname`, `nb_dots`, `nb_hyphens`, `nb_slash`, `nb_at`, `nb_qm`, `nb_and`, `nb_eq`, `nb_underscore`, `nb_percent`, `nb_colon`, `nb_www`, `nb_com`, `ratio_digits_url`, `ratio_digits_host`, `https_token`, `shortening_service`, `punycode`, `port`, `prefix_suffix`, `random_domain`, `ip`, `abnormal_subdomain`, `nb_subdomains`, `tld_in_path`, `tld_in_subdomain`, `statistical_report`, `suspecious_tld`, `phish_hints`.

3.2 Word / Token Features

Derived from splitting the URL and page content into tokens: `length_words_raw`, `char_repeat`, `shortest_words_raw`, `shortest_word_host`, `shortest_word_path`, `longest_words_raw`, `longest_word_host`, `longest_word_path`, `avg_words_raw`, `avg_word_host`, `avg_word_path`.

3.3 HTML Content Features

Extracted from the page DOM: `nb_hyperlinks`, `ratio_intHyperlinks`, `ratio_extHyperlinks`, `ratio_nullHyperlinks`, `nb_extCSS`, `ratio_intRedirection`, `ratio_extRedirection`, `ratio_intErrors`, `ratio_extErrors`, `login_form`, `external_favicon`, `links_in_tags`, `submit_email`, `ratio_intMedia`, `ratio_extMedia`, `sfh`, `iframe`, `popup_window`, `safe_anchor`, `onmouseover`, `right_clic`, `empty_title`, `domain_in_title`, `domain_with_copyright`.

3.4 External Service / WHOIS Features

From third-party reputation sources: `whois_registered_domain`, `domain_registration_length`, `domain_age`, `web_traffic`, `dns_record`, `google_index`, `page_rank`, `domain_in_brand`, `brand_in_subdomain`, `brand_in_path`.

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4. Descriptive Statistics — Numerical Features

Feature	Mean	Std	Min	25%	Median	75%	Max	Skewness
`length_url`	61.1	55.3	12	33	47	71	1641	8.09
`length_hostname`	21.1	10.8	4	15	19	24	214	5.16
`nb_dots`	2.48	1.37	1	2	2	3	24	5.72
`nb_hyphens`	1.00	2.09	0	0	0	1	43	4.70
`nb_slash`	4.29	1.88	2	3	4	5	33	2.73
`nb_subdomains`	2.23	0.64	1	2	2	3	3	−0.24
`ratio_digits_url`	0.053	0.089	0	0	0	0.079	0.724	2.19
`ratio_digits_host`	0.025	0.093	0	0	0	0	0.800	5.59
`nb_hyperlinks`	87.2	166.8	0	9	34	101	4659	7.68
`ratio_intHyperlinks`	0.602	0.376	0	0.225	0.743	0.945	1.0	−0.53
`ratio_extHyperlinks`	0.277	0.320	0	0	0.131	0.475	1.0	1.01
`page_rank`	3.19	2.54	0	1	3	5	10	0.45
`web_traffic`	856,757	1,995,606	0	0	1,651	373,846	10,767,990	2.78

Feature	Mean	Std	Min	25%	Median	75%	Max	Skewness
`domain_age`	4,063	3,108	−12	972	3,993	7,027	12,874	0.16
`domain_registration_length`	492.5	814.8	−1	84	242	449	29,829	9.82
`longest_words_raw`	15.4	22.1	2	9	11	16	829	13.53

Key observations:

- Most URL-lexical features are **heavily right-skewed** (skewness > 2), indicating a long tail of unusually long or complex URLs predominantly associated with phishing.
- `page_rank` and `domain_age` are the most normally distributed among all numerical features.
- Negative minimum values in `domain_age` (−12 days) and `domain_registration_length` (−1) are data anomalies and should be treated as noise or sentinel values during modelling.

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5. Missing Values

No missing values were found across all 89 columns. The dataset is complete and requires no imputation.

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6. Zero-Variance (Constant) Features

The following **6 features carry no information** as they take a single constant value across all 11,430 rows:

Feature	Constant Value	Implication
`nb_or`	0	Always zero — drop before modelling
`ratio_nullHyperlinks`	0	Always zero — drop before modelling
`ratio_intRedirection`	0	Always zero — drop before modelling
`ratio_intErrors`	0	Always zero — drop before modelling
`submit_email`	0	Always zero — drop before modelling
`sfh`	0	Always zero — drop before modelling

These features should be **removed** in preprocessing as they contribute zero discriminative power.

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7. Outlier Analysis (IQR Method)

Outliers were identified using the 1.5×IQR rule:

Feature	Outlier Count	% of Dataset
`length_hostname`	775	6.8%
`length_url`	620	5.4%
`nb_dots`	567	5.0%
`nb_slash`	401	3.5%
`page_rank`	0	0.0%
`domain_age`	0	0.0%

Insight: Outliers in URL-length features are not random errors — they represent **genuinely abnormal URLs** (e.g., obfuscated phishing links with hundreds of characters). These should be **retained** rather than removed, as they carry strong predictive signal. Tree-based models (Random Forest, XGBoost) are naturally robust to such outliers.

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8. Correlation Analysis

Top Correlations with Target (phishing = 1)

Feature	r with Target	Direction
`page_rank`	0.511	Higher rank → legitimate
`ratio_digits_url`	0.356	More digits → phishing
`nb_hyperlinks`	0.343	More links → phishing
`domain_age`	0.332	Older domain → legitimate
`length_url`	0.249	Longer URL → phishing
`ratio_intHyperlinks`	0.244	More internal links → legitimate
`nb_slash`	0.242	More slashes → phishing
`length_hostname`	0.238	Longer hostname → phishing
`ratio_digits_host`	0.224	Digits in host → phishing
`nb_dots`	0.207	More dots → phishing

Insight: `page_rank` is the single strongest correlate — phishing pages consistently have low or zero page rank since they are newly created and not indexed. `domain_age` and `web_traffic` reinforce this: phishing domains are typically new and have negligible traffic.

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9. Binary Feature Analysis

Key binary features and their proportion of value=1 per class:

Feature	Legitimate (prop=1)	Phishing (prop=1)	Δ Difference
`domain_in_title`	0.88	0.66	**−0.22** (legit higher)
`domain_with_copyright`	0.55	0.33	−0.22
`external_favicon`	0.36	0.52	+0.16
`https_token`	0.69	0.53	−0.16
`google_index`	0.60	0.47	−0.13
`empty_title`	0.09	0.16	+0.07
`login_form`	0.05	0.08	+0.03
`shortening_service`	0.13	0.12	~0

Insight:

- Legitimate pages are much more likely to include the domain in the page title and to have copyright notices — both indicators of an established brand identity.

- Phishing pages more frequently use external favicons (loading favicons from a trusted brand's server to appear legitimate).
- `shortening_service` shows almost no difference between classes, making it a weak discriminator.

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10. Feature Distribution Insights (Histograms)

- **`length_url`:** Legitimate URLs cluster tightly around 30–60 characters. Phishing URLs have a much wider, flatter distribution with a heavy tail extending past 200+ characters.
- **`nb_dots`:** Phishing URLs often embed multiple subdomains (e.g., `secure.bank.evil.com`) leading to more dots.
- **`nb_hyphens`:** Phishing URLs frequently use hyphens to mimic trusted domains (e.g., `paypal-login.com`).
- **`ratio_digits_url`:** Near-zero for legitimate URLs; phishing URLs often contain random digit sequences.
- **`page_rank`:** Legitimate pages are heavily concentrated at ranks 3–8; phishing pages cluster at 0.
- **`domain_age`:** Phishing domains are significantly younger; legitimate sites have ages uniformly distributed over years.

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11. Pair Plot Observations

The pair plot of top URL features reveals:

- `page_rank` × `length_url` shows clear class separation — legitimate pages occupy the high-rank, short-URL quadrant.
- `ratio_digits_url` × `nb_dots` shows phishing sites trending toward higher values on both axes.
- `nb_hyphens` × `nb_dots` reveals a cluster of phishing URLs with high values on both dimensions (hyphenated multi-subdomain attacks).

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12. Key Insights Summary

Finding	Detail
Perfectly balanced dataset	No class imbalance; standard metrics apply
No missing values	Dataset is analysis-ready
6 zero-variance features	Must be dropped before modelling
Heavy right skew	URL length features need log transformation
~5–7% outliers in URL features	Meaningful signal — do not remove
Strongest predictor	`page_rank` (
Top phishing indicators	High `ratio_digits_url`, long URLs, high `nb_dots`, high `nb_hyphens`, low `page_rank`, low `domain_age`
Top legitimacy indicators	High `page_rank`, old `domain_age`, `domain_in_title`=1, `domain_with_copyright`=1
Near-useless features	`shortening_service`, `brand_in_subdomain`, `brand_in_path` (low class separation)

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13. Recommendations for Modelling (Week 3)

1. **Drop** the 6 zero-variance constant features.
2. **Apply log1p transformation** to highly skewed features: `length_url`, `nb_hyperlinks`, `web_traffic`, `domain_registration_length`, `longest_words_raw`.

3. **Handle negative values** in `domain_age` and `domain_registration_length` (likely −1 or −12 as sentinel codes for "unknown") — replace with 0 or a dedicated flag feature.
4. **Feature importance ranking** will likely echo correlation analysis: `page_rank`, `domain_age`, `web_traffic`, `ratio_digits_url`, `length_url` will be top contributors.
5. Consider **tree-based ensembles** (Random Forest, XGBoost) as first models given the mixture of binary/count/continuous features and outlier-heavy distributions.
6. The `url` column (raw string) could be leveraged for additional feature engineering (TF-IDF character n-grams) if a more advanced approach is desired.

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14. Visualizations

The following figures are referenced in this report:

Figure	Description
`fig1_target_dist.png`	Target class distribution bar chart
`fig2_histograms.png`	Overlapping histograms of 9 key numerical features by class
`fig3_heatmap.png`	Correlation heatmap of numerical features
`fig4_boxplots.png`	Boxplots showing outliers per class for 6 features
`fig5_binary_features.png`	Binary feature proportion comparison across classes
`fig6_pairplot.png`	Pair plot of top 5 URL features (1,000 sample)

Report prepared for Week 2 EDA Assignment — Phishing URL Detection | Dataset: 11,430 rows × 89 columns